Phase-1

```
Administrator: Windows PowerShell
Windows PowerShell
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Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\WINDOWS\system32> cd "C:\Users\aadhi\kitchen-app"
PS C:\Users\aadhi\kitchen-app> node -v
v20.20.1
PS C:\Users\aadhi\kitchen-app> npm -v
10.8.2
PS C:\Users\aadhi\kitchen-app> docker --version
>>
Docker version 29.2.1, build a5c7197
PS C:\Users\aadhi\kitchen-app> kubectl version --client
Client Version: v1.34.1
Kustomize Version: v5.7.1
PS C:\Users\aadhi\kitchen-app> git --version
git version 2.49.0.windows.1
```

```
PS C:\Users\aadhi\kitchen-app\api> cd api
>> npm start
cd : Cannot find path 'C:\Users\aadhi\kitchen-app\api\api' because it does not exist.
At line:1 char:1
+ cd api
+ ~~~~~~
    + CategoryInfo          : ObjectNotFound: (C:\Users\aadhi\kitchen-app\api\api:String) [Set-Location], ItemNotFound
   Exception
    + FullyQualifiedErrorId : PathNotFound,Microsoft.PowerShell.Commands.SetLocationCommand


> kitchen-api@1.0.0 start
> node server.js

Kitchen API listening on port 3000
```

```
localhost:3000/health
localhost:3000/health
Pretty-print
{"status":"ok","db":"connected"}
```

Phase-2

```
PS C:\Users\aadhi\kitchen-app\api> docker build -t kitchen-api:local ./api
[+] Building 0.0s (0/0)                                                    docker:desktop-linux
ERROR: failed to build: unable to prepare context: path "./api" not found
```

```
PS C:\Users\aadhi\kitchen-app\api> docker build -t kitchen-api:local .
[+] Building 36.0s (15/15) FINISHED                                                                    docker:desktop-linux
 => [internal] load build definition from Dockerfile                                                                    0.1s
 => => transferring dockerfile: 478B                                                                                    0.0s
 => [internal] load metadata for docker.io/library/node:20-alpine                                                       2.9s
 => [auth] library/node:pull token for registry-1.docker.io                                                             0.0s
 => [internal] load .dockerignore                                                                                       0.0s
 => => transferring context: 2B                                                                                         0.0s
 => [internal] load build context                                                                                       5.6s
 => => transferring context: 51.54MB                                                                                    5.3s
 => [builder 1/5] FROM docker.io/library/node:20-alpine@sha256:b88333c42c23fbd91596ebd7fd10de239cedab9617de04142d  0.0s
 => => resolve docker.io/library/node:20-alpine@sha256:b88333c42c23fbd91596ebd7fd10de239cedab9617de04142dde7315e3  0.0s
 => CACHED [production 2/5] RUN addgroup -g 1001 -S nodejs && adduser -S nodeuser -u 1001 -G nodejs                    0.0s
 => CACHED [production 3/5] WORKDIR /app                                                                                0.0s
 => CACHED [builder 2/5] WORKDIR /build                                                                                 0.0s
 => [builder 3/5] COPY package*.json ./                                                                                 0.3s
 => [builder 4/5] RUN npm ci && npm cache clean --force                                                                15.8s
 => [builder 5/5] COPY . .                                                                                              2.4s
 => [production 4/5] COPY --from=builder --chown=nodeuser:nodejs /build/node_modules ./node_modules                     1.7s
 => [production 5/5] COPY --chown=nodeuser:nodejs server.js package.json ./                                             0.2s
 => exporting to image                                                                                                  5.6s
 => => exporting layers                                                                                                 2.8s
 => => exporting manifest sha256:700115824477f92cc4ccd39b77ebeb402edb62235566b20c4edbcfd755c9d536                       0.0s
 => => exporting config sha256:0ca39fe0bca290b4f823538f0248b3c7f62402b7cfb7ede53f0adcac51b9f11d                         0.0s
 => => exporting attestation manifest sha256:cafb0b83e7893a83b219c8452ab9c6a52a20d41b990288c6e9503e420599a8bc           0.0s
 => => exporting manifest list sha256:60395c74e4e1201478080109ce18aaba287f21fe6c9971739abe73562cd860fd                  0.0s
 => => naming to docker.io/library/kitchen-api:local                                                                    0.0s
 => => unpacking to docker.io/library/kitchen-api:local                                                                 2.6s
PS C:\Users\aadhi\kitchen-app\api>
```

```
PS C:\Users\aadhi\kitchen-app\api> docker compose up -d
>> docker compose ps
[+] up 3/4
 ✔Network kitchen-app_default    Created                                                                                0.1s
 ✔Container kitchen-db           Healthy                                                                                9.0s
 - Container kitchen-app-api-1   Starting                                                                               8.7s
 ✔Container kitchen-app-web-ui-1 Created                                                                                0.2s
Error response from daemon: ports are not available: exposing port TCP 0.0.0.0:3000 -> 127.0.0.1:0: listen tcp 0.0.0.0:3000: bind: Only one usage of each socket address (protocol/network address/port) is normall
y permitted.
NAME          IMAGE       COMMAND                  SERVICE   CREATED         STATUS                    PORTS
kitchen-db    mongo:7.0   "docker-entrypoint.s…"   db        9 seconds ago   Up 8 seconds (healthy)    0.0.0.0:27017->27017/tcp, [::]:27017->27017/tcp
PS C:\Users\aadhi\kitchen-app\api>
```

Phase-3

```
PS C:\Users\aadhi\kitchen-app> kubectl get ns
NAME              STATUS   AGE
default           Active   10d
kitchen-app       Active   10d
kube-node-lease   Active   10d
kube-public       Active   10d
kube-system       Active   10d
PS C:\Users\aadhi\kitchen-app> kubectl get pods -n kitchen-app
NAME                                    READY   STATUS    RESTARTS      AGE
api-deployment-8687b75877-lv4qx         1/1     Running   1 (37m ago)   2d2h
api-deployment-8687b75877-lxhtc         1/1     Running   2 (26m ago)   2d2h
grafana-deployment-f9dcd87f9-47z95      1/1     Running   1 (37m ago)   2d
mongo-0                                 1/1     Running   1 (37m ago)   2d2h
mongodb-0                               1/1     Running   2 (37m ago)   10d
prometheus-deployment-66489fcbb5-8ck5q  1/1     Running   1 (37m ago)   2d1h
web-ui-deployment-5c9574dcc4-62gkc      1/1     Running   2 (26m ago)   46h
web-ui-deployment-5c9574dcc4-8x6z8      1/1     Running   2 (26m ago)   46h
PS C:\Users\aadhi\kitchen-app> kubectl get svc -n kitchen-app
NAME                TYPE        CLUSTER-IP       EXTERNAL-IP   PORT(S)          AGE
api                 ClusterIP   10.98.22.123     <none>        3000/TCP         10d
api-nodeport        NodePort    10.102.129.120   <none>        3000:30000/TCP   10d
grafana-service     NodePort    10.111.131.131   <none>        3000:30300/TCP   2d
mongo               ClusterIP   10.106.61.125    <none>        27017/TCP        2d2h
mongodb-service     ClusterIP   10.97.198.249    <none>        27017/TCP        10d
prometheus-service  NodePort    10.102.67.23     <none>        9090:30090/TCP   2d1h
web-ui-service      NodePort    10.106.91.99     <none>        80:30080/TCP     10d
PS C:\Users\aadhi\kitchen-app>
```

```
mongodb-0                          1/1     Running   0          2m21s
PS C:\Users\aadhi\kitchen-app> kubectl apply -f k8s/web-ui/deployment.yaml
error: the path "k8s/web-ui/deployment.yaml" does not exist
PS C:\Users\aadhi\kitchen-app> kubectl apply -f k8s/web-ui/deployment.yaml
deployment.apps/web-ui-deployment created
service/web-ui-service created
PS C:\Users\aadhi\kitchen-app> kubectl get pods -n kitchen-app
NAME                                READY   STATUS    RESTARTS   AGE
api-deployment-7f4cb5d99b-qpd9r     1/1     Running   0          11m
mongodb-0                           1/1     Running   0          13m
web-ui-deployment-544c496fc4-q8sn9  1/1     Running   0          96s
web-ui-deployment-544c496fc4-rvjsm  1/1     Running   0          96s
PS C:\Users\aadhi\kitchen-app> kubectl get svc -n kitchen-app
NAME              TYPE        CLUSTER-IP       EXTERNAL-IP   PORT(S)          AGE
api               ClusterIP   10.98.22.123     <none>        3000/TCP         12m
api-nodeport      NodePort    10.102.129.120   <none>        3000:30000/TCP   12m
mongodb-service   ClusterIP   10.97.198.249    <none>        27017/TCP        14m
web-ui-service    NodePort    10.106.91.99     <none>        80:30080/TCP     117s
PS C:\Users\aadhi\kitchen-app> Invoke-RestMethod http://localhost:30000/health

status db
------ --
ok     connected


PS C:\Users\aadhi\kitchen-app>
```

```
web-ui-deployment-544c496fc4-q8sn9  1/1     Running   0          96s
web-ui-deployment-544c496fc4-rvjsm  1/1     Running   0          96s
PS C:\Users\aadhi\kitchen-app> kubectl get svc -n kitchen-app
NAME              TYPE        CLUSTER-IP       EXTERNAL-IP   PORT(S)          AGE
api               ClusterIP   10.98.22.123     <none>        3000/TCP         12m
api-nodeport      NodePort    10.102.129.120   <none>        3000:30000/TCP   12m
mongodb-service   ClusterIP   10.97.198.249    <none>        27017/TCP        14m
web-ui-service    NodePort    10.106.91.99     <none>        80:30080/TCP     117s
PS C:\Users\aadhi\kitchen-app> Invoke-RestMethod http://localhost:30000/health

status db
------ --
ok     connected


PS C:\Users\aadhi\kitchen-app> Invoke-RestMethod -Method Post -Uri "http://localhost:30000/orders" -ContentType "applic
ation/json" -Body (@{ dish = "Chicken Biryani" } | ConvertTo-Json)


_id             : 69b531930f3b9cc2fa1bcd4f
dish            : Chicken Biryani
status          : pending
createdAt       : 2026-03-14T09:59:47.742Z
statusUpdatedAt : 2026-03-14T09:59:47.742Z
nextStatus      : preparing
nextStatusAt    : 2026-03-14T09:59:52.742Z


PS C:\Users\aadhi\kitchen-app>
```

2. Click Create order

Phase-4

```
Administrator: Windows PowerShell
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Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\WINDOWS\system32> cd "C:\Users\aadhi\kitchen-app"
PS C:\Users\aadhi\kitchen-app> kubectl apply -f k8s/namespace.yaml
namespace/kitchen-app unchanged
PS C:\Users\aadhi\kitchen-app> kubectl get ns
NAME              STATUS   AGE
default           Active   8d
kitchen-app       Active   8d
kube-node-lease   Active   8d
kube-public       Active   8d
kube-system       Active   8d
PS C:\Users\aadhi\kitchen-app> kubectl create configmap api-config --from-literal=PORT=3000 --namespace kitchen-app --dr
y-run=client -o yaml | kubectl apply -f -
>>
>> kubectl create configmap api-server-override --from-file=server.js=api/server.js --namespace kitchen-app --dry-run=cl
ient -o yaml | kubectl apply -f -
>>
>> kubectl create secret generic mongodb-secret `
>> --from-literal=mongo-root-username=admin `
>> --from-literal=mongo-root-password=Secret123! `
>> --from-literal=mongo-uri='mongodb://admin:Secret123!@mongodb-service.kitchen-app.svc.cluster.local:27017/kitchendb?au
thSource=admin' `
>> --namespace kitchen-app --dry-run=client -o yaml | kubectl apply -f -
configmap/api-config created
configmap/api-server-override created
secret/mongodb-secret created
PS C:\Users\aadhi\kitchen-app>
```
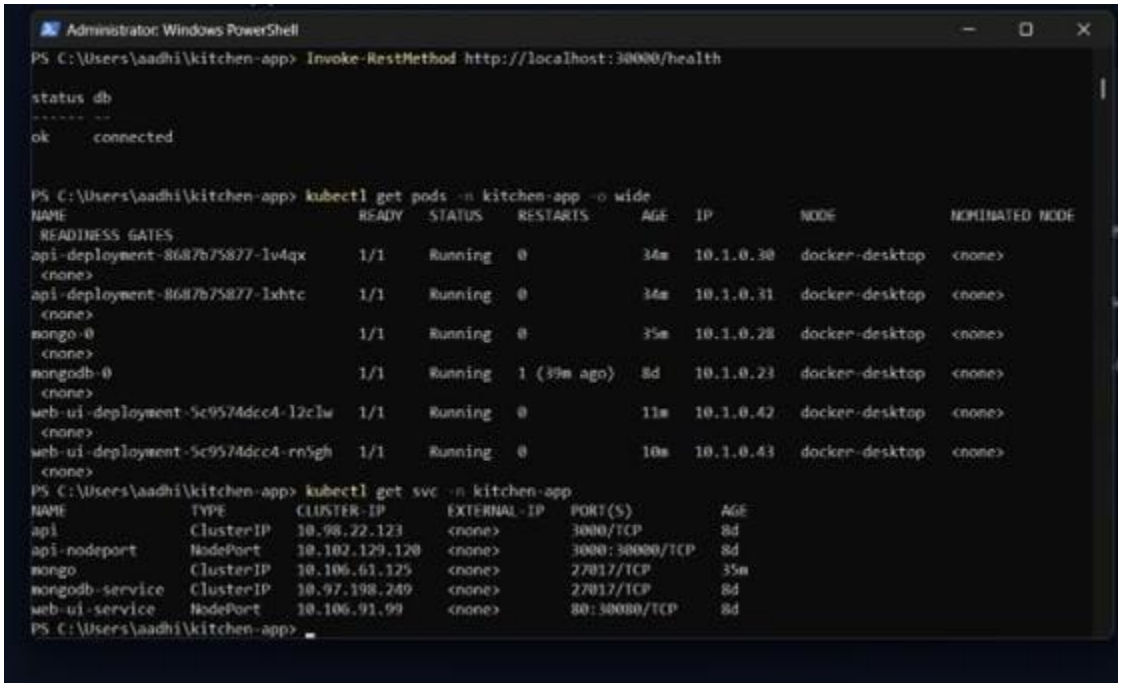
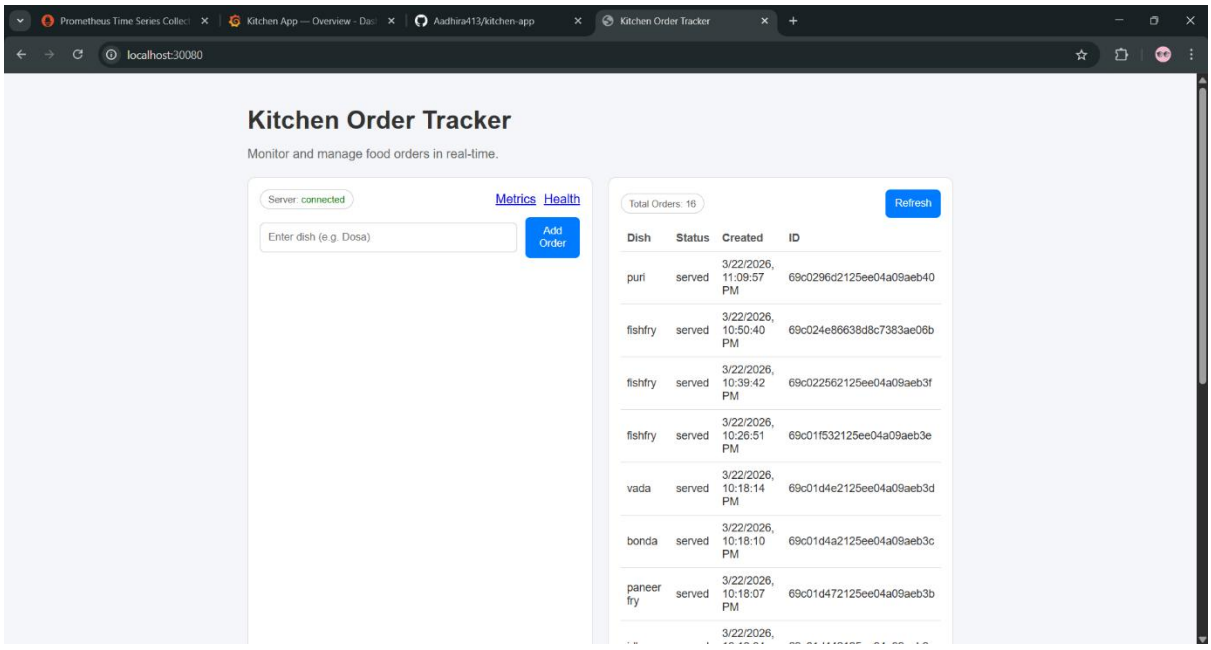

```
Administrator: Windows PowerShell
Waiting for deployment "api-deployment" rollout to finish: 1 old replicas are pending termination...
Waiting for deployment "api-deployment" rollout to finish: 1 old replicas are pending termination...
deployment "api-deployment" successfully rolled out
PS C:\Users\aadhi\kitchen-app> kubectl apply -f k8s/web-ui/deployment.yaml
configmap/web-ui-config created
deployment.apps/web-ui-deployment configured
service/web-ui-service unchanged
PS C:\Users\aadhi\kitchen-app> kubectl rollout status deployment/web-ui-deployment -n kitchen-app --timeout=90s
Waiting for deployment "web-ui-deployment" rollout to finish: 1 out of 2 new replicas have been updated...
Waiting for deployment "web-ui-deployment" rollout to finish: 1 out of 2 new replicas have been updated...
Waiting for deployment "web-ui-deployment" rollout to finish: 1 out of 2 new replicas have been updated...
Waiting for deployment "web-ui-deployment" rollout to finish: 1 old replicas are pending termination...
Waiting for deployment "web-ui-deployment" rollout to finish: 1 old replicas are pending termination...
deployment "web-ui-deployment" successfully rolled out
PS C:\Users\aadhi\kitchen-app> kubectl get pods -n kitchen-app -o wide
NAME                                READY   STATUS    RESTARTS        AGE     IP          NODE             NOMINATED NO
DE   READINESS GATES
api-deployment-8687b75877-lv4qx     1/1     Running   0               2m23s   10.1.0.30   docker-desktop   <none>
     <none>
api-deployment-8687b75877-lxhtc     1/1     Running   0               2m8s    10.1.0.31   docker-desktop   <none>
     <none>
mongo-0                             1/1     Running   0               3m3s    10.1.0.28   docker-desktop   <none>
     <none>
mongodb-0                           1/1     Running   1 (7m18s ago)   8d      10.1.0.23   docker-desktop   <none>
     <none>
web-ui-deployment-5958759b99-bzgnq  1/1     Running   0               44s     10.1.0.33   docker-desktop   <none>
     <none>
web-ui-deployment-5958759b99-dn48r  1/1     Running   0               66s     10.1.0.32   docker-desktop   <none>
     <none>
PS C:\Users\aadhi\kitchen-app>
```



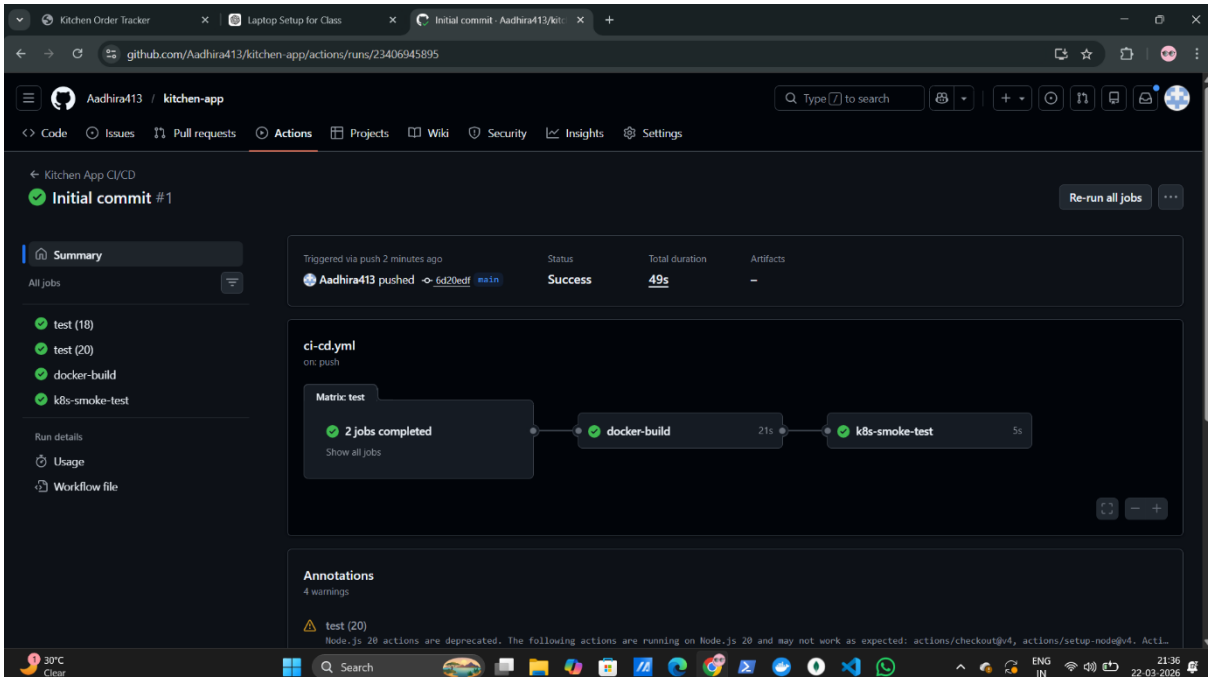
```
Administrator: Windows PowerShell
PS C:\Users\aadhi\kitchen-app> Invoke-RestMethod http://localhost:30000/health

status db
------ --
ok     connected


PS C:\Users\aadhi\kitchen-app> kubectl get pods -n kitchen-app -o wide
NAME                                  READY   STATUS    RESTARTS      AGE   IP          NODE             NOMINATED NODE
 READINESS GATES
api-deployment-8687b75877-lv4qx       1/1     Running   0             34m   10.1.0.30   docker-desktop   <none>
 <none>
api-deployment-8687b75877-lxhtc       1/1     Running   0             34m   10.1.0.31   docker-desktop   <none>
 <none>
mongo-0                               1/1     Running   0             35m   10.1.0.28   docker-desktop   <none>
 <none>
mongodb-0                             1/1     Running   1 (39m ago)   8d    10.1.0.23   docker-desktop   <none>
 <none>
web-ui-deployment-5c9574dcc4-l2clw    1/1     Running   0             11m   10.1.0.42   docker-desktop   <none>
 <none>
web-ui-deployment-5c9574dcc4-rn5gh    1/1     Running   0             10m   10.1.0.43   docker-desktop   <none>
 <none>
PS C:\Users\aadhi\kitchen-app> kubectl get svc -n kitchen-app
NAME              TYPE        CLUSTER-IP       EXTERNAL-IP   PORT(S)          AGE
api               ClusterIP   10.98.22.123     <none>        3000/TCP         8d
api-nodeport      NodePort    10.102.129.120   <none>        3000:30000/TCP   8d
mongo             ClusterIP   10.106.61.125    <none>        27017/TCP        35m
mongodb-service   ClusterIP   10.97.198.249    <none>        27017/TCP        8d
web-ui-service    NodePort    10.106.91.99     <none>        80:30080/TCP     8d
PS C:\Users\aadhi\kitchen-app>
```



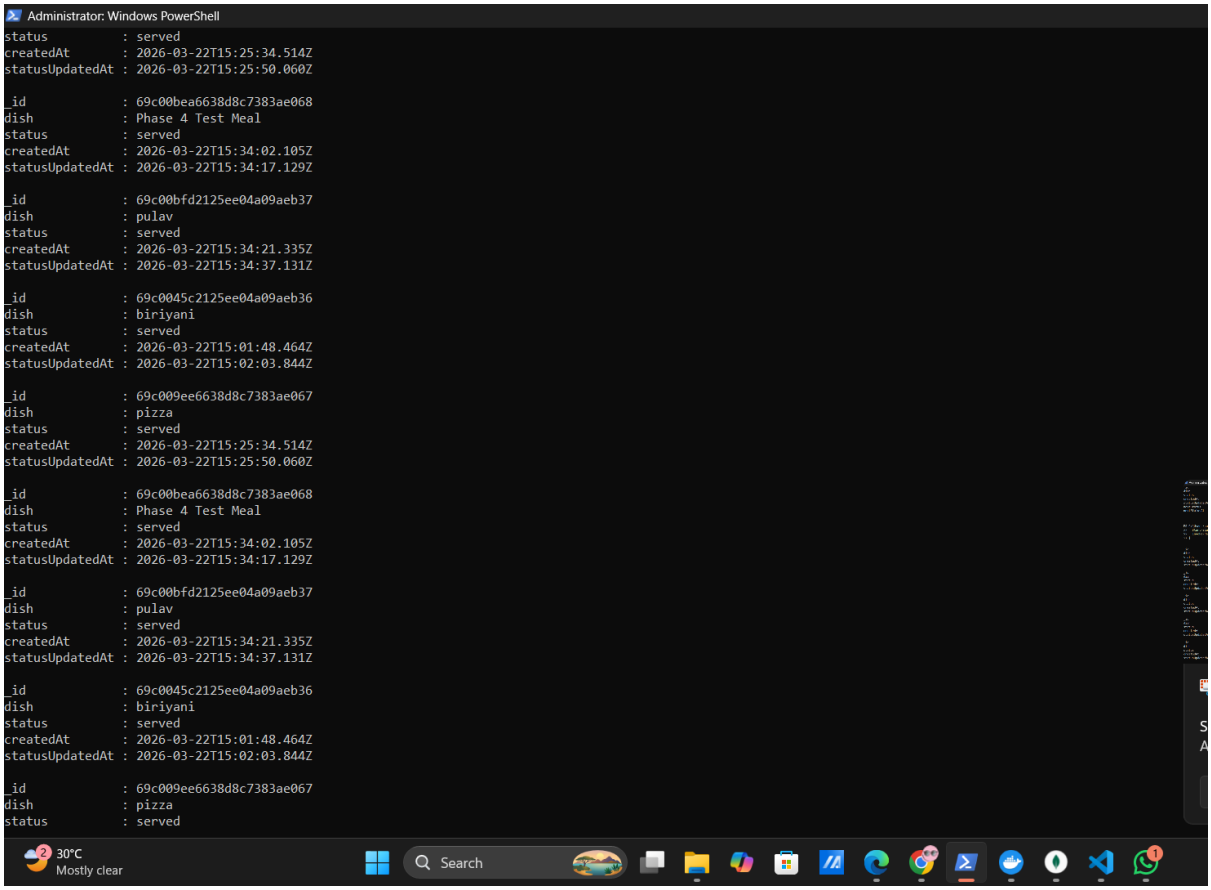
Phase-5

```
PS C:\Users\aadhi\kitchen-app> git commit -m "Initial commit"
[master (root-commit) 6d20edf] Initial commit
 20 files changed, 6767 insertions(+)
 create mode 100644 .github/workflows/ci-cd.yml
 create mode 100644 .gitignore
 create mode 100644 api/Dockerfile
 create mode 100644 api/eslint.config.mjs
 create mode 100644 api/package-lock.json
 create mode 100644 api/package.json
 create mode 100644 api/server.js
 create mode 100644 api/server.test.js
 create mode 100644 docker-compose.yml
 create mode 100644 k8s/api/deployment.yaml
 create mode 100644 k8s/api/service.yaml
 create mode 100644 k8s/db/statefulset.yaml
 create mode 100644 k8s/monitoring/grafana-dashboard.yaml
 create mode 100644 k8s/monitoring/grafana-deployment.yaml
 create mode 100644 k8s/monitoring/grafana-provisioning.yaml
 create mode 100644 k8s/monitoring/prometheus-config.yaml
 create mode 100644 k8s/namespace.yaml
 create mode 100644 k8s/web-ui/deployment.yaml
 create mode 100644 web-ui/Dockerfile
 create mode 100644 web-ui/index.html
```

```
nothing to commit, working tree clean
PS C:\Users\aadhi\kitchen-app> git branch -M main
PS C:\Users\aadhi\kitchen-app> git remote add origin https://github.com/Aadhira413/kitchen-app.git
error: remote origin already exists.
PS C:\Users\aadhi\kitchen-app> git remote remove origin
PS C:\Users\aadhi\kitchen-app> git remote add origin https://github.com/Aadhira413/kitchen-app.git
PS C:\Users\aadhi\kitchen-app> git remote -v
origin  https://github.com/Aadhira413/kitchen-app.git (fetch)
origin  https://github.com/Aadhira413/kitchen-app.git (push)
PS C:\Users\aadhi\kitchen-app> git push -u origin main
Enumerating objects: 31, done.
Counting objects: 100% (31/31), done.
Delta compression using up to 12 threads
Compressing objects: 100% (27/27), done.
Writing objects: 100% (31/31), 61.04 KiB | 3.21 MiB/s, done.
Total 31 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), done.
To https://github.com/Aadhira413/kitchen-app.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
PS C:\Users\aadhi\kitchen-app>
sistance
```


Kitchen Order Tracker
Laptop Setup for Class
Initial commit - Aadhira413/kit
github.com/Aadhira413/kitchen-app/actions/runs/23406945895
Aadhira413 / kitchen-app
Type / to search
Code
Issues
Pull requests
Actions
Projects
Wiki
Security
Insights
Settings
Kitchen App CI/CD
Initial commit #1
Re-run all jobs
Summary
All jobs
test (18)
test (20)
docker-build
k8s-smoke-test
Run details
Usage
Workflow file
Triggered via push 2 minutes ago
Aadhira413 pushed 6d20edf main
Status
Success
Total duration
49s
Artifacts
–
ci-cd.yml
on: push
Matrix: test
2 jobs completed
Show all jobs
docker-build 21s
k8s-smoke-test 5s
Annotations
4 warnings
test (20)
Node.js 20 actions are deprecated. The following actions are running on Node.js 20 and may not work as expected: actions/checkout@v4, actions/setup-node@v4. Acti…
30°C
Clear
Search
ENG
IN
21:36
22-03-2026

Phase-6


Administrator: Windows PowerShell
status : served
createdAt : 2026-03-22T15:25:34.514Z
statusUpdatedAt : 2026-03-22T15:25:50.060Z
_id : 69c00bea6638d8c7383ae068
dish : Phase 4 Test Meal
status : served
createdAt : 2026-03-22T15:34:02.105Z
statusUpdatedAt : 2026-03-22T15:34:17.129Z
_id : 69c00bfd2125ee04a09aeb37
dish : pulav
status : served
createdAt : 2026-03-22T15:34:21.335Z
statusUpdatedAt : 2026-03-22T15:34:37.131Z
_id : 69c0045c2125ee04a09aeb36
dish : biriyani
status : served
createdAt : 2026-03-22T15:01:48.464Z
statusUpdatedAt : 2026-03-22T15:02:03.844Z
_id : 69c009ee6638d8c7383ae067
dish : pizza
status : served
createdAt : 2026-03-22T15:25:34.514Z
statusUpdatedAt : 2026-03-22T15:25:50.060Z
_id : 69c00bea6638d8c7383ae068
dish : Phase 4 Test Meal
status : served
createdAt : 2026-03-22T15:34:02.105Z
statusUpdatedAt : 2026-03-22T15:34:17.129Z
_id : 69c00bfd2125ee04a09aeb37
dish : pulav
status : served
createdAt : 2026-03-22T15:34:21.335Z
statusUpdatedAt : 2026-03-22T15:34:37.131Z
_id : 69c0045c2125ee04a09aeb36
dish : biriyani
status : served
createdAt : 2026-03-22T15:01:48.464Z
statusUpdatedAt : 2026-03-22T15:02:03.844Z
_id : 69c009ee6638d8c7383ae067
dish : pizza
status : served
30°C
Mostly clear
Search

```
Administrator: Windows PowerShell
status          : served
createdAt       : 2026-03-22T15:25:34.514Z
statusUpdatedAt : 2026-03-22T15:25:50.060Z

_id             : 69c00bea6638d8c7383ae068
dish            : Phase 4 Test Meal
status          : served
createdAt       : 2026-03-22T15:34:02.105Z
statusUpdatedAt : 2026-03-22T15:34:17.129Z

_id             : 69c00bfd2125ee04a09aeb37
dish            : pulav
status          : served
createdAt       : 2026-03-22T15:34:21.335Z
statusUpdatedAt : 2026-03-22T15:34:37.131Z


PS C:\Users\aadhi\kitchen-app>
```

```
PS C:\Users\aadhi\kitchen-app> kubectl rollout status deployment/prometheus-deployment -n kitchen-app --timeout=120s
deployment "prometheus-deployment" successfully rolled out
PS C:\Users\aadhi\kitchen-app> kubectl apply -f k8s/monitoring/grafana-dashboard.yaml
configmap/grafana-dashboards created
PS C:\Users\aadhi\kitchen-app> kubectl apply -f k8s/monitoring/grafana-deployment.yaml
deployment.apps/grafana-deployment created
service/grafana-service created
PS C:\Users\aadhi\kitchen-app> kubectl rollout status deployment/grafana-deployment -n kitchen-app --timeout=120s
Waiting for deployment "grafana-deployment" rollout to finish: 0 of 1 updated replicas are available...
error: timed out waiting for the condition
PS C:\Users\aadhi\kitchen-app> kubectl get pods -n kitchen-app
NAME                                    READY   STATUS              RESTARTS      AGE
api-deployment-8687b75877-lv4qx         1/1     Running             0             76m
api-deployment-8687b75877-lxhtc         1/1     Running             0             76m
grafana-deployment-7bcd9bd94c-4qntc     0/1     ContainerCreating   0             5m23s
mongo-0                                 1/1     Running             0             77m
mongodb-0                               1/1     Running             1 (81m ago)   8d
prometheus-deployment-66489fcbb5-8ck5q  1/1     Running             0             32m
web-ui-deployment-5c9574dcc4-l2clw      1/1     Running             0             52m
web-ui-deployment-5c9574dcc4-rn5gh      1/1     Running             0             52m
PS C:\Users\aadhi\kitchen-app>
```

```
Administrator: Windows PowerShell
indows PowerShell
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nstall the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

S C:\WINDOWS\system32> cd "C:\Users\aadhi\kitchen-app"
S C:\Users\aadhi\kitchen-app> kubectl port-forward -n kitchen-app svc/prometheus-service 9090:9090
orwarding from 127.0.0.1:9090 -> 9090
orwarding from [::1]:9090 -> 9090
andling connection for 9090
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```